APMM Web Retrieval

API Sources and Cost Breakdown | TruthLens Capstone | Sep 2026

Overview

The pipeline uses three source categories: image reverse search (finds where the input image appears across the web), social media crawling (finds posts on closed platforms), and news/open web (indexed news coverage). All prices are in AUD at 1 USD = 1.55 AUD.

Category 1 — Image Reverse Search

Google Cloud Vision — Web Detection

Primary reverse image search. Sends the image to Google's index and returns up to 100 pages where it appears. More comprehensive than SerpAPI as it hits Google's index directly with no middleman.

Pricing: pay as you go, per image. No monthly commitment.

Volume	Cost AUD / 1,000 calls
First 1,000 / month	Free
1,001 to 5,000,000 / month	~$5.43
Over 5,000,001 / month	Contact Google

Cost at our volumes: free for testing and research (under 1,000 calls/month). ~$22 AUD at scale (5,000 cases).

SerpAPI — Google Lens

Secondary reverse image search. Wraps Google Lens — returns slightly different results to Cloud Vision, stronger on product matches and exact copies. Running both and deduplicating on URL gives better combined coverage. At scale, Cloud Vision is cheaper and more direct so SerpAPI becomes optional.

Pricing: monthly subscription tiers. No pay-as-you-go option. Searches do not roll over.

Plan	Searches / month	Cost AUD / month
Free (permanent)	250	Free
Starter	1,000	~$39
Developer	5,000	~$116
Production	15,000	~$233
Big Data	50,000	~$426

Category 2 — Social Media Crawling

X / Twitter API v2

Finds tweets and posts containing or referencing the image. Critical for tracking social media propagation. X is typically where viral images first amplify before spreading to news outlets. Pay-per-use is the only model available to new developers since February 2026 — monthly tiers are closed to new signups.

Pricing: pay as you go per tweet read. Cap of 2,000,000 reads/month.

Action	Cost AUD per unit
Reading others' tweets	~$0.0078
Reading your own tweets	~$0.0016

Estimated at 100 tweet reads per image case:

Stage	Cases / month	Tweet reads	Cost AUD
Testing	5	500	~$4
Research	50	5,000	~$39
Scale (20 users)	500	50,000	~$388

X/Twitter dominates cost at scale. Mitigation: only pull Twitter data for cases where the image is already visually confirmed by Vision Web Detection, reducing reads by ~60-70%.

Instagram and Facebook — Apify

Finds public Instagram posts and Facebook public page posts. Apify provides pre-built scrapers that handle bot detection, proxy rotation, and CAPTCHA — significantly more reliable than DIY Playwright scraping. Remarkably cheap at research and testing volumes.

Pricing: credit-based. Each plan includes monthly prepaid usage credits at $0.20 per compute unit. Unused credits expire each cycle. Free plan requires no credit card.

Plan	Monthly Cost AUD	Credits (page checks)
Free (permanent)	Free	~500 checks ($5 credit)
Starter	~$45	~2,900 checks
Scale	~$308	~20,000 checks

Instagram and Facebook — Playwright Sessions (Alternative)

Log in manually once, save the session cookies, reuse in the crawler at no ongoing cost. Sessions expire after days to weeks and must be refreshed. Good for testing and one-off analysis. Not suitable for high-frequency heartbeat monitoring.

Cost: $0 at all volumes.

Meta Content Library and API (Future)

Official Meta research tool providing structured access to public Facebook and Instagram content including images, share counts, and timestamps. Far more comprehensive than scraping. Requires application at researchtools.meta.com. UQ affiliation and misinformation research purpose qualify. Approval takes 2 to 6 weeks.

Cost: $0 once approved.

Category 3 — News and Open Web

GDELT + RSS Feeds

GDELT indexes global news media coverage worldwide across 100+ languages. Used to find articles referencing an image's story by keyword. RSS feeds subscribe directly to major outlets including Reuters, AP, BBC, NBC, and Al

Jazeera. Both are used in the heartbeat loop to monitor for new coverage.

Cost: $0 at all volumes. No API key required.

Monthly Cost Summary

Testing Phase — right now, minimal cases

Source	Purpose	Cost AUD / month
Google Cloud Vision	Primary reverse image search	Free
SerpAPI (free tier)	Secondary reverse image search	Free
X / Twitter API	Social media propagation	~$4
Apify (free tier)	Instagram + Facebook crawling	Free
GDELT + RSS	News coverage	Free
TOTAL		~$4

Research Phase — grant budget, 50 cases/month

Source	Purpose	Cost AUD / month
Google Cloud Vision	Primary reverse image search	Free
SerpAPI Starter	Secondary reverse image search	~$39
X / Twitter API	Social media propagation	~$39
Apify (free tier)	Instagram + Facebook crawling	Free
GDELT + RSS	News coverage	Free
TOTAL		~$78

Scale Phase — 20 users, 500 cases/month

Source	Purpose	Cost AUD / month
Google Cloud Vision	Primary reverse image search	~$22
SerpAPI Developer	Secondary reverse image search	~$116
X / Twitter API	Social media propagation	~$388 (~$140 selective)
Apify Starter	Instagram + Facebook crawling	~$45
GDELT + RSS	News coverage	Free
TOTAL (full)		~$571
TOTAL (selective Twitter)		~$323

Platforms Not Currently Covered

Platform	Access Route	Status
Facebook + Instagram (full)	Meta Content Library	Application required, 2-6 weeks
TikTok	TikTok Research API	Application required
WhatsApp	No public API exists	Not possible
LinkedIn	Partner API only	Not available for this use case